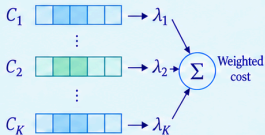


Penalty view

Costs enter the objective



Exact settings

Graph-price cap = 0

Adaptive β : feedback only

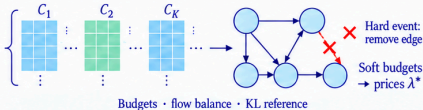
Frozen β : fixed penalty

SafeOT-Dual

One price rule, two views

Joint costs + budgets

One graph problem



Clip + filter



Realized-cost feedback

Realized episode cost $\bar{J}_k - B_k \rightarrow$ accumulate

$\bar{\lambda}^*$

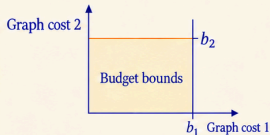
β



Shared prices λ

Constraint view

Budgets constrain graph costs



Under assumptions

Conditional graph-price limits

$\varepsilon \rightarrow 0$: LP prices

$\varepsilon \rightarrow \infty +$ clip: active-set saturation

One policy-update rule

Priced advantage

$$\frac{A_r - \sum_k \lambda_k A_{ck}}{1 + \sum_k \lambda_k}$$

PPO / TRPO

$\pi(a | s)$